

Transfer Learning and Permutation-Invariance Improving Predicting Genome-Wide, Cell-Specific and Directional Interventions Effects of Complex Systems

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With the rise of precision medicine, single-drug treatments alone may not meet its demands. However, extensive data and deep learning models exist for single compounds. Leveraging transfer learning to use this data for predicting complex system intervention effects is promising. In this study, a deep model based on permutation-invariance is used as the core module, pre-trained on a large amount of single-compound intervention data in cell lines, and fine-tuned on a small amount of complex system (like natural products) intervention data in cell lines, resulting in a predictive model, Set Embedding and Transfer learning model for Complex systems (SETComp). The two versions of SETComp achieved an accuracy of 93.86% and 92.70%, respectively, on the complex system-cell-gene association test set, improving by 5.82% to 27.59% compared to the baseline. When predicting the intervention effects of those complex systems the model has never encountered before, the accuracy increased by up to 24.83% compared to the baseline. In the in vitro validation, up to 88.65% of the predictions are confirmed to be correct, and the model's output showed a significant positive correlation with the real-world foldchange. SETComp's potential in various biomedical scenarios is further observed, achieving good performance in applications such as mechanism uncovering and drug repositioning.

1. Introduction

With the advancement of artificial intelligence and high-throughput omics technologies, researchers have gained a deeper understanding of life science or biomedical issues such as disease mechanisms and drug development. It has gradually become evident that single drugs may not adequately address the complex mechanisms underlying complex diseases. In this context, combination therapies involving multiple drugs or treatments based on complex systems like natural products (NP) have gradually become more popular, with predicting the targets of multiple drugs or complex systems with high throughput and high accuracy emerging as a hot topic in biomedicine. However, compared to target prediction for single compounds,^[1–5] target prediction for complex systems requires more consideration on how to integrate compound–compound interactions, compound side effects and other information. Additionally, compared to target prediction for combinations of two or three compounds,^[6–8] predicting targets for complex systems like NP, which involve an

indeterminate number of compounds, appears more challenging. Currently, there are few existing algorithms for target prediction of complex systems, particularly network-based approaches. Li et al.^[9,10] proposed statistical modeling based on the targets of all compounds to predict the holistic targets of complex systems. Zhou et al.^[11–13] treated complex systems as whole entities and utilized heterogeneous graph neural networks for prediction. However, these prediction methods have certain limitations, including the inability to make predictions based on specific entities such as target cells, low prediction accuracy, no ability for directional prediction and a reliance on extensive prior knowledge, such as the need for the targets of compounds contained within NP. Therefore, developing a cell-specific target prediction model and a target promotion/inhibition direction prediction model suitable for complex systems with a large number of compounds is highly valuable and holds great potential for biomedical application.

Transfer learning, as a deep learning technique,^[14,15] has excellent applicability in scenarios with limited sample sizes. Till

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now, there is a large amount of data accumulated on compound-intervened transcriptomics like Connectivity Map (CMap) for The Library of Integrated Network-Based Cellular Signatures (LINCS) project.^[16] In comparison, transcriptomic data related to complex systems is relatively scarce in comparison. Therefore, applying transfer learning to complex systems-related applications is highly meaningful. Similarly, set-based deep learning possesses the property of permutation invariance, with Deep Sets^[17] and Set Transformer^[18] being classic models in this domain. Set-based deep learning methods are designed to handle unordered data, such as sets, by ensuring that the model's output remains invariant regardless of the input data's order. This property makes them particularly effective for tasks where the order of elements in the set is irrelevant, such as in complex systems with multiple compounds. These models provide a powerful approach to understanding and predicting the interactions and effects of various components within these systems.

In this study, we employed transfer learning techniques by treating complex systems as sets of compound combinations and utilizing permutation-invariant (set-based) deep learning as the core model framework, named Set Embedding and Transfer learning model for Complex systems (SETComp), to predict genome-wide, cell-specific and directional targets for both compounds and complex systems. We pre-trained the model on 970481750 compound–cell–gene association data obtained from transcriptomics data processed from LINCS and further fine-tuned it on 2579488 natural product–cell–gene association data collected and processed from GEO datasets and literature. Two versions of SETComp (Concat version with $\approx 200\text{M}$ parameters and Add version with $\approx 173\text{M}$ parameters) achieved 93.86% and 92.70% accuracy in predicting NP-cell-gene associations, respectively, of which the performance was also tested in real-world in vitro assays, achieving high accuracy. Besides, in multiple downstream application scenarios, SETComp has demonstrated good performance, including revealing the potential mechanisms of action of complex systems on different cell lines corresponding to tumors and repositioning complex systems to discover new potential diseases. In summary, we have trained a model based on set-based deep learning using vast amounts of compound-cell-gene data and applied transfer learning to achieve high-precision predictions of complex system-cell-gene associations, which demonstrates strong application potential in various downstream scenarios related to biomedical issues.

2. Results

2.1. Overview of the Study

In this study, we proposed a model named Set Embedding and Transfer learning model for Complex systems (SETComp), cored by the combination of transfer learning and set-based deep learning. Aiming at predicting transcriptome-level targets in different types of cells, SETComp was designed for stimulating the genome-wide changes in the cellular level after the intervening of complex compound systems like drug combinations or NP.

To achieve modeling the effects at the transcriptomic level after the intervention of a single compound or complex systems like NP in different cell conditions (Figure S1, Supporting Informa-

tion), SETComp consists of three frozen encoders and three internal modules for feature exacting, feature integration and performing prediction (Figure 1). The three frozen encoders are: a compound encoder, which combines the trained Infograph graph neural network and a Fingerprinter encoder; a gene encoder, integrating a lightweight pre-trained large language model (LLM) with a node2vec model derived from protein-protein interaction (PPI) networks; and a cell state encoder, which is a Variational Auto-Encoders (VAE) model trained on expression profiles from The Cancer Genome Atlas (TCGA). As for the internal modules, centered around the set embedding module, which consists of an initial embedding based on Deep Sets^[17] and a further embedding based on Set Transformer,^[18] the three internal modules also include a self-supervised module and a prediction module composed of multiple progressively dimension-reduced Multi-layer Perceptron (MLP).

Compounds with SMILES description, genes, and cell lines were embedded into feature space with the three frozen encoders, respectively. SETComp was first pre-trained on 970481750 compound–cell–gene association data obtained from 1805898 transcriptomics samples with a single intervention of 39321 compounds, to achieve accurate prediction of transcriptomics-level changes. With the collected relationships between compounds and NP, every natural product was considered as a set of compounds, and the model was then fine-tuned on 2579488 natural product-cell-gene pairs extracted from collected data obtained from both the GEO database (update to 2024 May) and literature for predicting transcriptomics-level targets for complex compound systems like NP. In addition, in vitro experimental validations were conducted on different cell lines and NPs to estimate the performance of SETComp in real-world assays. In the downstream, SETComp has been tested for applications in multiple biomedical-related scenarios, including the analysis of intervention mechanisms at the molecular and pathway levels, as well as drug repositioning.

2.2. Pre-Training of the SETComp Model

Although RNA-seq technology has been around for over a decade and has accumulated a large amount of data in the context of single compound interventions, such as the CMap from LINCS project, data in the case of complex system interventions, such as NP interventions, is relatively lacking compared to single compounds. We have collected transcriptomic data from compound interventions in CMap which then enhanced by a modified CycleGAN,^[19] transcriptomic data from NP interventions from GEO and literature (Figure 2a), original transcriptomics data from TCGA and CCLE, gene association networks and protein coding information from STRING,^[20] compound structure information from PubChem,^[21] and compound composition information for NP from HERB (v2.0)^[22] (Figure S2a, Supporting Information). After cleaning, screening, enhancement, and preprocessing the transcriptomic data of single compound interventions, we obtained gene differential expression data for over 20000 compounds (Figure S2b, Supporting Information) across more than 90 cell lines (Figure S2c, Supporting Information) and constructed 970481750 compound-cell line-gene association pairs.

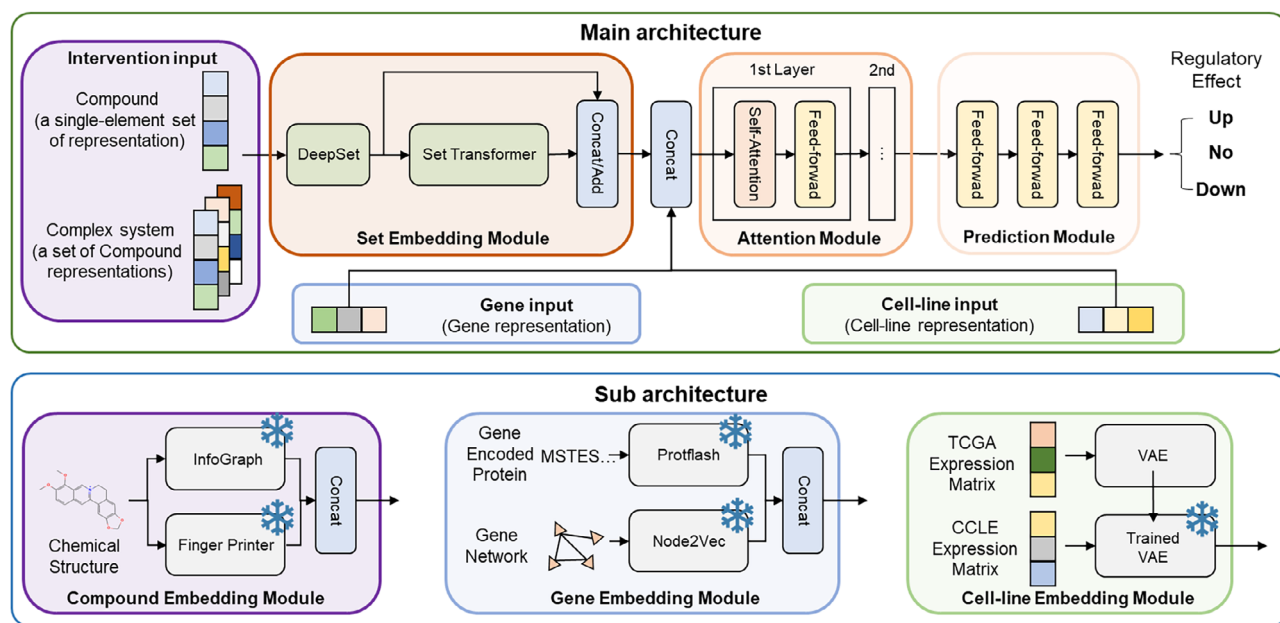


Figure 1. Model schematic of the SETComp model. Architecture of SETComp. The core module is the permutation-invariant Set Embedding Module, which characterizes the complex system as a set, with initial and deep characterizations via Deep Sets and Set Transformer. The set-based embedding is combined with gene and cell line representations, followed by a self-attention module to learn feature relationships. The prediction module classifies associations as upregulation, downregulation, or no significant change. Compounds are represented by the self-supervised pre-trained InfoGraph and Fingerprinter models, while genes are represented by PPI-based node2vec and the pre-trained Protflash model. Cell lines are represented by the latent layer of the trained Variational Auto-Encoders (VAE). All submodule models are frozen during feature extraction.

The enhanced transcriptomic expression data of untreated cell lines showed a discernible data distribution in UMAP space, indicating that the data-enhanced cell lines possess improved representational characteristics, better clustering, and more robust differentiation in gene expression patterns (Figure 2b; Figure S2d, Supporting Information). The core module of the SETComp model, the Set embedding module (Figure S3, Supporting Information), consisted of two set-based deep learning model, Deep Sets^[17] and Set Transformer^[18] (see supplementary materials). Deep sets initially characterize the input sets (including single-element sets composed of individual compounds and multi-element sets composed of complex systems) as fixed-length sets, providing a preliminary representation of the input. The output after Deep sets representation is then used as a new set input, which is fed into the set Transformer. Through its unique Multi-head Attention Block (MAB), Set Attention Block (SAB), Induced Set Attention Block (ISAB), and Pooling by Multihead Attention (PMA) blocks, further deep representation is obtained. Meanwhile, we employ two strategies, concatenation and addition, to combine the shallow set-based representation from Deep sets and the deep set-based representation obtained from the collaborative characterization of Deep sets and the set Transformer by either concatenating or adding them, to construct the Concat version ($\approx 200\text{M}$ parameters) and the Add version ($\approx 173\text{M}$ parameters) of the SETComp model. After concatenating the output of the set embedding module with the gene features and the cell line features obtained from the VAE, self-attention is applied to further learn the relationships between the features. Finally, the set-cell-gene prediction is performed through a prediction module composed of three layers of MLP with progressively reduced

dimensions. After performing a grid search on the training parameters, including model parameters such as the size of the set Transformer, the size of MLP in the prediction module, as well as training parameters like batch size, learning rate, and regularization parameters such as L2 regularization and dropout, the optimal model parameters for the Concat version were used to determine the model structure (Table S1, Supporting Information). Subsequently, the Add version also underwent a grid search on the training and regularization parameters (Table S2, Supporting Information). Both versions of the SETComp model approach convergence after 5 epochs (Tables S3 and S4, Supporting Information), and without overfitting, we trained both versions of the model to 10 epochs with the optimal parameters.

In our training set division for pre-training, all compounds belonging to NP components were extracted separately as the test set. By training on the entire training set, we found that the vanilla neural network composed only of the prediction module outperforms the two versions of the SETComp model (with AUCs of 0.9480 and 0.9417, respectively) in predicting compound-cell-gene associations, and further outperforms the model (the NoSet version), which consists of the self-attention module and the prediction module (Figure S4a–d; Table S5, Supporting Information). This may be because when predicting a single compound, treating it as an individual rather than as a set is more suitable and sufficiently saturated for the model. The two versions of the SETComp model perform better in predicting up-regulation and down-regulation, with an AUC of 0.96 for both (Figure S5a–d, Supporting Information). Further, on a smaller-scale training set (1/100 size of the full training set), we found that deep learning-based models performed better on the test set

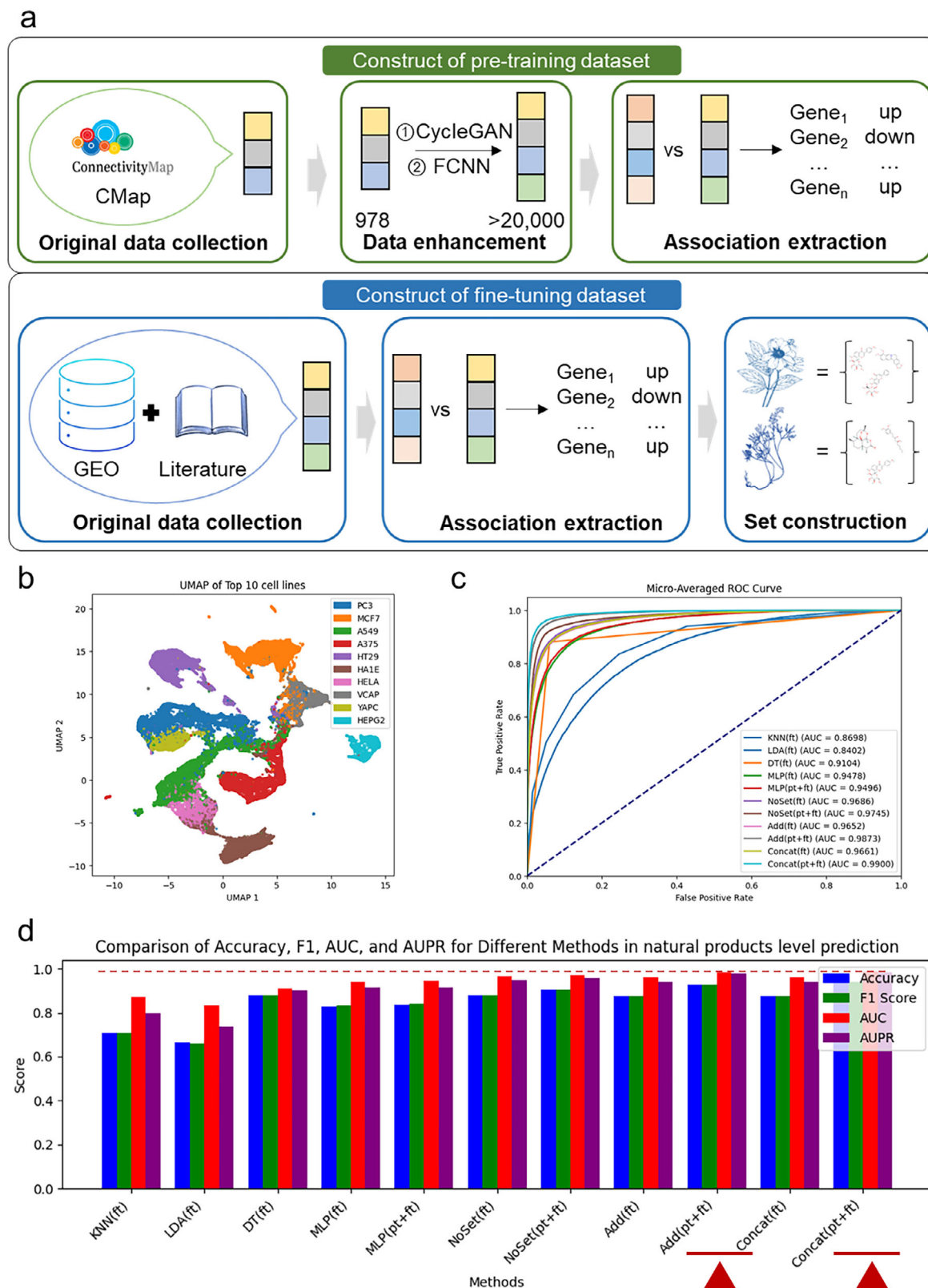


Figure 2. Preprocessing and performance of the SETComp model. a) Dataset construction for SETComp. Compound-intervened expression profiles from multiple cell lines for pre-training were obtained from CMap in the LINCS project. These profiles were enhanced using a modified CycleGAN model to 978-D RNA-seq-like profiles and then extrapolated to 23614-D whole-genome profiles with a trained full connected neural network. Compound-cell-gene associations were derived from differential expression analysis. Fine-tuning used intervention profiles from complex systems (e.g., natural

compared to traditional machine learning models (Figure S6a–d; Table S6, Supporting Information), including K-Nearest Neighbors (KNN), Linear Discriminant Analysis (LDA), and Decision Trees (DT).

2.3. Fine-Tuning Improving the Performance of Prediction on Complex Systems

After completing the model pre-training, we focused on our research goal—achieving high-precision complex system-cell-gene association prediction. We pre-processed various complex system sets with compound features as elements, with the largest set containing up to 406 elements (Figure S7a, Supporting Information). At the same time, we cleaned, pre-processed, and analyzed expression data of complex systems from the GEO database and literature across different cell lines (Figure S7b,c, Supporting Information). We found that there was no significant correlation between the total expression levels of cell lines after complex system interventions, such as NP, and the number of compounds (Figure S7d, Supporting Information) contained in NP ($P = 0.4331$).

During the fine-tuning procedure, we also performed a grid search on the training parameters, including batch size and learning rate, as well as regularization parameters such as L2 regularization and dropout for the two versions of the SETComp model, to determine the optimal parameter combination (Table S7, Supporting Information). We found that the performance of the two versions of the SETComp model, after pre-training and fine-tuning, surpassed traditional deep learning models such as KNN, LDA, and DT, as well as the vanilla neural network and the NoSet version in deep learning, and their performance when trained only on fine-tuning (Figure 2c). The two versions of the SETComp model achieved 93.86% and 92.70% in accuracy (Table S8, Supporting Information), as well as 0.9888 and 0.9856 in AUC, respectively, improving the accuracy compared to the baseline machine learning model, increasing from 5.82% to 27.59%, while the AUC increases from 7.83% to 15.63% (Figure S8a,c, Supporting Information) and the AUPR increases from 8.20% to 24.4% (Figure S8b, Supporting Information). Compared to the vanilla neural network, the accuracy can be improved by 10.10%, and compared to the NoSet version, it shows an improvement of 5.74%.

Compared to the model that only underwent pre-training, we found that on the test set, the model that underwent both pre-training and fine-tuning performed better than the model that only underwent fine-tuning, and both outperformed the model that only underwent pre-training. This finding also demonstrates the benefit of fine-tuning on the prediction model, as well as the effectiveness of pre-training on compound-cell-gene in enhancing the model's performance (Figure 3a; Figure S9a–c, Table

S9, Supporting Information). At the same time, this finding also indicates that, beyond the superiority of deep learning itself in this task, the attention module and set embedding module progressively contribute to the improvement of the model's performance. Then, based on the pre-trained model, we gradually added the size of the fine-tuning training set from 0% to 100%, and observed that as the size of the fine-tuning training set increased, the model's performance on the fine-tuning test set gradually improved for both the Concat version (Figure 3b; Figure S10a,b, Supporting Information) and the Add version (Figure S10c, Supporting Information) of the SETComp model. Furthermore, by adding models pre-trained on a small-scale pre-training dataset, we also performed fine-tuning and found that a larger pre-training dataset led to better performance on the fine-tuning test set, while accelerating the improvement of Accuracy and AUC (Figure 3c; Figure S11c,d, Supporting Information) as well as the reduction of loss (Figure S11a,b, Supporting Information), and improving Accuracy and AUC of the converged models (Figure S12a–d, Table S10, Supporting Information).

Furthermore, to assess the model's generalization ability, we re-divided the fine-tuning dataset based on NP, using 10% of the NP-related transcriptomic data as the test set, and the remaining data as the training and test sets for fine-tuning the model, and re-trained the pre-trained model. When predicting unseen NP, it can achieve accuracies of 82.75% and 82.66% for the two versions of the model (Figure 3d; Table S11, Supporting Information), respectively, which is an improvement of up to 24.83% in accuracy compared to the baseline machine learning models and 5.59% to the vanilla neural network (Figure S13a–c, Supporting Information). Finally, to explore the potential for quantitative prediction in the future, we analyzed the model output values (predicted as down-regulated if negative) after applying softmax on all NP-cell-gene associations in the fine-tuning test set, and their actual statistical test logFoldChange (Figure 3e). We found a significant correlation, and this correlation varied across different NPs, ranging from 0.6734 to 0.9226 (Figure S14, Supporting Information). This finding suggests that the model's predicted results have a significant positive correlation with the actual transcriptomic logFoldChange, and it somewhat indicates the feasibility of extending the model to quantitative prediction.

2.4. In Vitro Experiments Validating the Performance of the SETComp Model

To further validate the predictive performance of the model, we conducted real-world transcriptomics assays to verify the model's predictions, especially in unseen NP-cell lines combinations. In multiple cell lines, we tested the effects of several NPs, including cinnamon, codonopsis, and astragalus (Figure 4a), and obtained

products) across multiple cell lines, obtained from GEO datasets and literature. Complex system-cell-gene associations were similarly processed based on differential expression. The sets for each complex system were constructed using recorded composition data from HERB (v2.0). b) UMAP visualization of the embeddings of the top 10 cell lines with the largest number of interventions in the pre-training dataset, after PCA dimensionality reduction. c) ROC curves of fine-tuned SETComp and baseline models (including K-Nearest Neighbors (KNN), Linear Discriminant Analysis (LDA), Decision Trees (DT), a vanilla neural network (MLP), and the NoSet version consisting of only the attention and prediction modules) on the test set. d) Performance comparison of fine-tuned SETComp and baseline models on the test set. Both the Concat and Add versions of SETComp outperformed traditional machine learning models, vanilla neural networks, and the NoSet version in accuracy, F1 score, AUC, and AUPR.

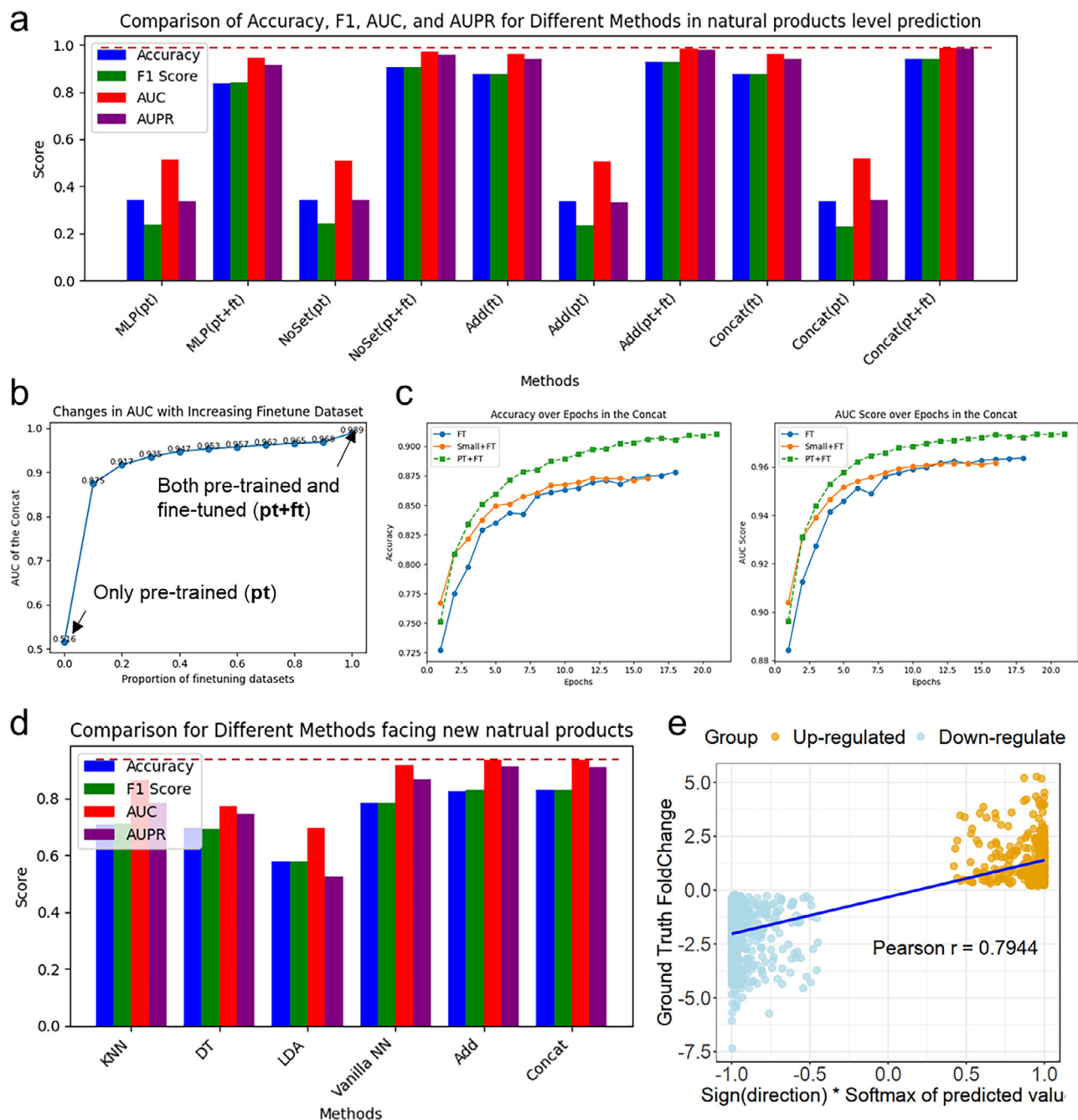


Figure 3. Ablation experiments and generalization study of the SETComp model. a) Performance comparison of fine-tuned SETComp and other deep learning models on the test set, trained with different datasets. Both the Concat and Add versions of SETComp, after pre-training and fine-tuning, achieved the highest performance in accuracy, F1 score, AUC, and AUPR, surpassing the vanilla neural network and NoSet version. These models also outperformed their respective performance when trained solely on pre-training or fine-tuning data. b) Line chart showing how the AUC score of the Concat version of SETComp changes with the increasing percentage of the fine-tuning dataset. As the fine-tuning dataset size increases, the model's AUC also improves. c) Line chart showing the variation in Accuracy and AUC scores of the Concat version of SETComp with different pre-training conditions. Models trained with the full pre-training dataset outperformed those pre-trained on smaller datasets or without pre-training in terms of both Accuracy and AUC. d) Performance of SETComp and baseline models on the test set extracted from unseen complex system intervention data. SETComp (Concat and Add versions) outperformed traditional machine learning models and vanilla neural networks in accuracy, F1 score, AUC, and AUPR. e) The relationship between the model's predicted output and Foldchange in real-world data. After softmax transformation, the model's predictions for class 0 (up-regulated) and class 1 (down-regulated) show a significant positive correlation with real-world data's Foldchange.

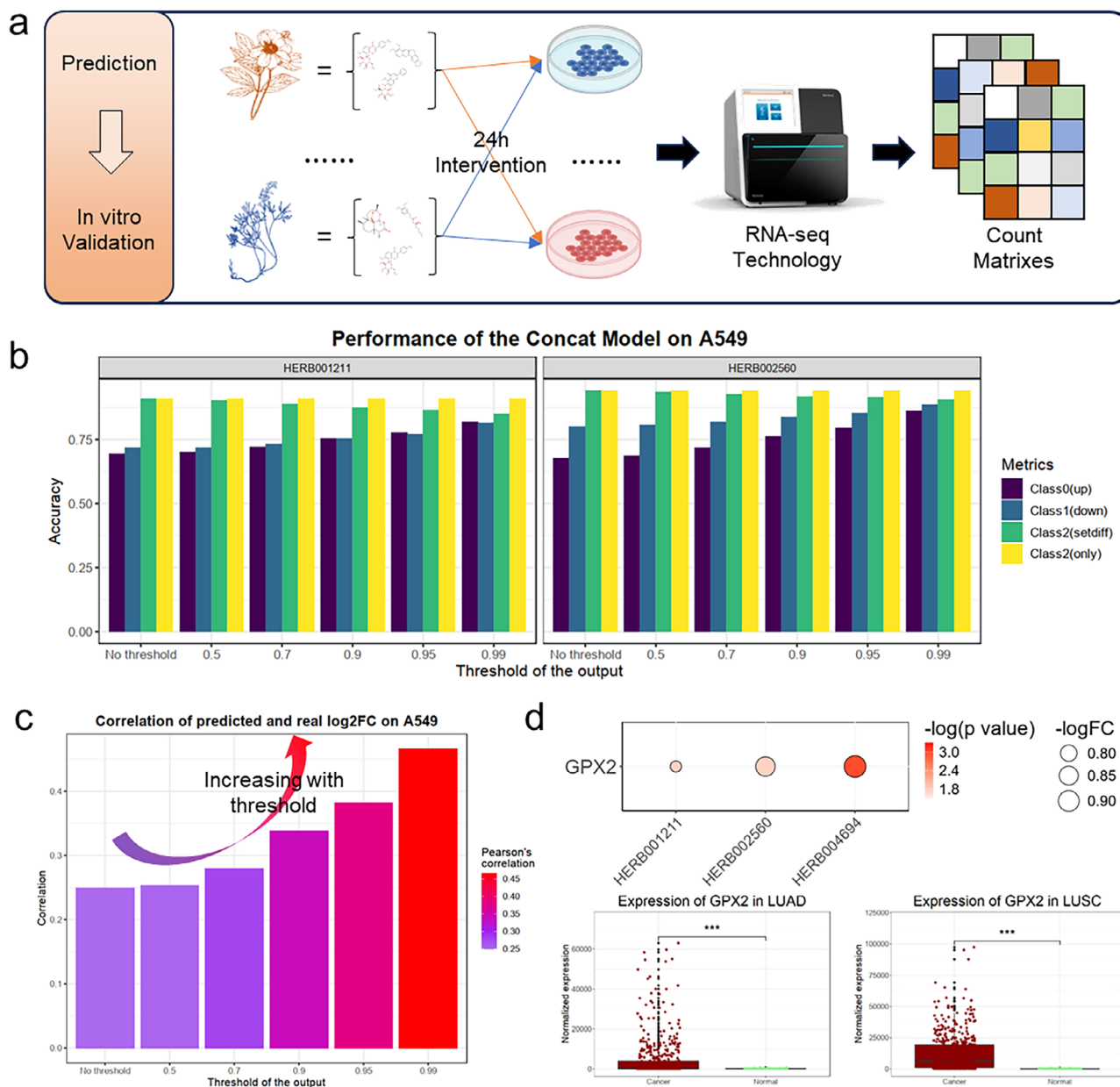


Figure 4. Case study of the SETComp model. a) Case study illustration. We conducted 24-h interventions of multiple complex systems (natural products) on cell lines and extracted RNA for RNA sequencing. b) Accuracy of SETComp's prediction of the intervention effects of Astragalus and Codonopsis in the A549 cell line within the differential expressed genes. In the A549 cell line, the accuracy of SETComp's prediction of the intervention effects of Astragalus and Codonopsis for class 0 and 1, increases as the threshold increases, reaching a maximum of 88.65%. For the prediction of class 2, the accuracy can reach up to 94.01%. c) The correlation coefficient between the SETComp predicted scores and the logFC obtained from real-world experiments within the differential expressed genes. As the threshold increases, the correlation coefficient between the SETComp predicted scores and the logFC from real-world experiments gradually increases, reaching a maximum of 0.4665. d) Potential target GPX2 identified by combining model predictions, real transcriptomic data, and TCGA tumor data. After integrating the model's predicted results (threshold 0.7) with transcriptomic intervention results, it was found that the three NPs collectively downregulated GPX2 and many other genes. Among them, six genes were found to be significantly overexpressed in tumor tissues of LUAD and LUSC. The upper bubble plot shows the downregulation of GPX2 in A549 cells after treatment with three NP in real-world transcriptomics assay.

their RNA-seq counts through transcriptome sequencing for differential expression analysis.

After conventional bioinformatics processing (Figure S15a, Supporting Information), we obtained the differential expression of each gene after cell line intervention. Among the differential

genes (p value < 0.05), for the model-predicted upregulated targets of codonopsis in A549, 75.44% were found to be truly upregulated (with a threshold of 0.9), while 75.28% of the downregulated targets were confirmed to be downregulated; for the model-predicted upregulated targets of astragalus in A549, 76.25% were

found to be truly upregulated (with the threshold of 0.9), and 83.95% of the downregulated targets were confirmed to be downregulated (Figure 4b). These accuracies improve as the threshold increases, reaching up to 81.72% for the upregulated prediction and 81.51% for the downregulated prediction for codonopsis at the threshold of 0.99; for astragalus, the upregulated prediction reaches 86.25% and the downregulated prediction reaches 88.65%. For cinnamon, the maximum accuracy in predicting downregulation can reach 83.12% (Figure S15b, Supporting Information). Among all genes, in the prediction of upregulated genes, the accuracy under the intervention of codonopsis or astragalus can reach a maximum of 76.84% (Figure S15c, Supporting Information), with accuracy showing an increasing trend as the threshold increases. In the prediction of negative samples (class 2), the model's predictions remain relatively stable, with a maximum of 94.01% of predictions for class samples being confirmed as having no significant differences for astragalus, and 90.85% for codonopsis.

Similarly, we also observed the relationship between the scores predicted by the model for each gene under different NP interventions after softmax and the actual fold change of gene expression under each NP intervention. We found that the expression of intervention genes predicted by the model (classified as 0 or 1) is consistently significantly positively correlated with their true fold change, and as the threshold increases, the correlation score between them gradually increases (Figure 4c). In the differential genes of the A549 cell line, this correlation can reach a maximum of 0.4665, and in all genes, it can reach 0.2409 (Figure S16a, Supporting Information). Focusing on the results of individual NP interventions, the positive correlation coefficient remains significant, and this phenomenon of increasing positive correlation with increasing threshold is still maintained. In the differential genes of the A549 cell line, the regularization coefficients under the interventions of astragalus, codonopsis, and cinnamon can reach 0.5788, 0.5486, and 0.4425, respectively (Figure S16b, Supporting Information); in the full gene range of the A549 cell line, the regularization coefficients under the interventions of astragalus, codonopsis, and cinnamon can reach 0.2306, 0.2146, and 0.2778, respectively (Figure S16c, Supporting Information). This finding further demonstrates that the model's predictions not only enable qualitative directional predictions and predictions of intervention, but also possess the potential for quantitative prediction of relative expression levels, providing evidence support for subsequent quantitative predictive analyses in application scenarios.

Combining the prediction results of the SETComp model of multiple complex systems on A549, real-world transcriptomics assays and transcriptomics data from TCGA, we attempted to identify potential targets of some complex systems in the treatment of the A549-related clinical cancers, specifically non-small cell lung cancer LUAD and LUSC. Among the genes that were predicted by the model with high scores (threshold of 0.7), that genes showed significant differential expression in the transcriptome, and those were significantly different in both LUAD and LUSC, we found that some potential targets such as GPX2 (Figure 4d), PRR13, and APOC1, which are highly expressed in cancer samples of LUAD and LUSC (Figure S17b,c, Supporting Information), were significantly downregulated under the intervention of all these three NP (Figure S17a, Supporting Informa-

tion). GPX2 has been studied in lung cancer, with reports indicating its involvement in apoptosis,^[23,24] immune regulation^[25] and oxidative stress,^[26–28] with clinical significance^[29] in lung cancer. The above findings, along with relevant literature, provide some evidence that SETComp's predictive capability demonstrates strong generalization, robustness, and potential for application expansion.

2.5. The Extensive Applications of the SETComp Model in Various Biomedical Issues

Last but not least, based on the model validated in both the test set and real-world assays, we use multiple biomedical application scenarios as downstream tasks to explore the practical application value of the model. First, we attempted to apply SETComp to explore the molecular and pathway mechanisms of complex system interventions in cell lines.

Based on the aforementioned finding, that the model's predicted values quantitatively reflect the activation/inhibition intensity of genes after intervention to some extent, we can apply Gene Set Enrichment Analysis (GSEA) to perform enrichment analysis on the prediction results and observe the potential pathway-level changes they may induce. We predicted the intervention effects of three NPs, including cinnamon, codonopsis, and astragalus (Figure S18a, Supporting Information) on two cell lines, MCF-7 and A549, and constructed their potential regulatory molecular networks (Figure S18b–d, Supporting Information). After GSEA based on the predicted up-regulated and down-regulated genes, astragalus was found to affect certain pathways in multiple modules in both MCF-7 and A549 cell lines (Figure 5a,b, Supporting Information). Although astragalus potentially intervenes in pathways belonging to modules such as signal transduction, cell growth and death, and the immune system in both MCF-7 and A549 cell lines, the specific pathways within each module vary. For example, in cell growth and death, astragalus potentially regulates tumor cell growth and proliferation in MCF-7 by promoting the p53 signaling pathway,^[30–32] while in the A549 cell line, it potentially promotes tumor cell apoptosis by enhancing the Apoptosis signaling pathway.^[33–35] Additionally, in some classic signaling pathways, astragalus potentially inhibits tumor cell growth and proliferation in MCF-7 through the FOXO signaling pathway,^[36–38] while in A549-related non-small cell lung cancer, it potentially alters the tumor immune microenvironment by promoting the Toll-like receptor signaling pathway.^[39,40]

Similar analysis also revealed that in the potential pathways of cinnamon and codonopsis interventions in MCF-7 (Figure S19a,b, Supporting Information) and A549 (Figure S20a,b, Supporting Information), the model is able to identify different potential molecules, pathways, and modular mechanisms for different complex systems intervening in different cell lines. Overall, more than 85% of the pathways were also significantly enriched in the transcriptome data ($p < 0.05$). The mechanistic discoveries at the pathway level align with our results from the transcriptomics data, suggesting the appropriate application of SETComp in the analysis of complex system interventions. Additionally, we observed the impact of different thresholds on the robustness of the analysis results. Taking the intervention of astragalus in MCF-7 and A549 as an example, we found that when the thresh-

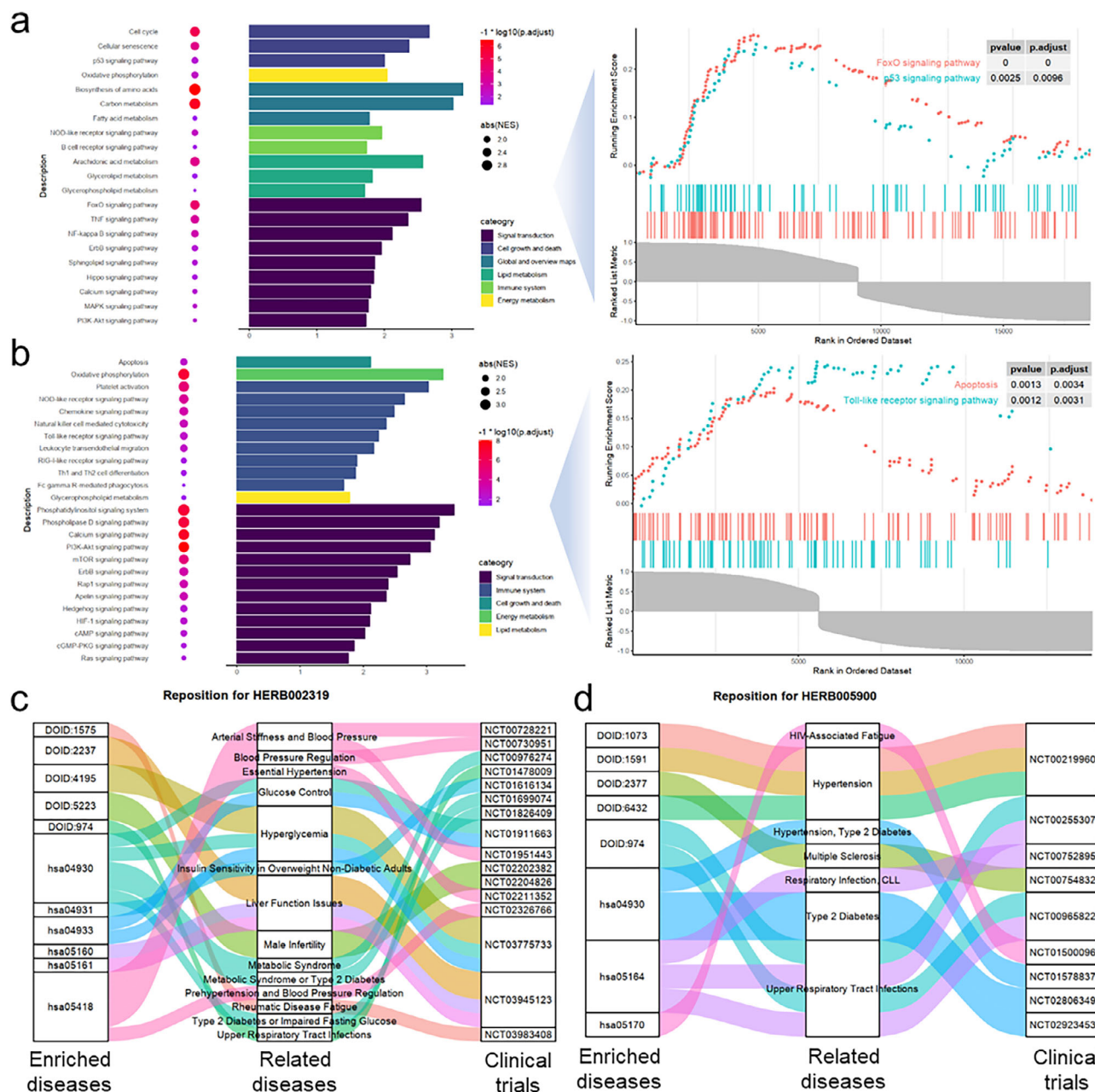


Figure 5. Application of SETComp in various biomedical scenarios. a,b) Mechanism uncovering of potential intervention pathways of Astragalus in MCF-7 (a) and A549 cell lines (b) based on the prediction of SETComp. The GSEA enrichment analysis revealed that SETComp predicted Astragalus to be involved in multiple modules in both MCF-7 and A549 cell lines, like signal transduction, cell growth and death. However, the specific effects on MCF-7 and A549 cells differ. For example, Astragalus primarily affects the p53 signaling pathway in MCF-7, whereas in A549, it mainly potentially upregulates the apoptosis signaling pathway. c,d) Reposition of certain complex systems like Red Ginseng (c) and American Ginseng (d) based on the prediction of SETComp. KEGG and DO enrichment analysis based on SETComp prediction results revealed that Red Ginseng and American Ginseng have potential interventions for multiple diseases, and this was supported by evidence from NCT clinical trials.

old was set to 0.7 (Figure S21a,b, Supporting Information) or 0.9 (Figure S21c,d, Supporting Information), the analysis results did not change significantly. The variations were likely observed in the same pathways but with different enrichment levels.

On the other hand, we explored the potential of SETComp in the field of drug repositioning. We analyzed several NPs with substantial clinical reports, including ginseng, American ginseng,

red ginseng, and cinnamon, and, after masking the clinical reports in advance, we observed whether the model could predict repositioning with clinical evidence. We first performed Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis and Disease Ontology (DO) analysis on the model's output. Based on different KEGG disease classifications, we organized the potential diseases that these NPs may intervene in (Figure S22a-d,

Supporting Information). According to the predicted KEGG and DO disease terms, red ginseng was found to liver function-related issues like hepatitis (DOID:1575, hsa05160, hsa05161) validated by clinical trial NCT0395412, and certain glucose-related diseases like Type II diabetes mellitus (hsa04930), Insulin resistance (hsa04931) and hyperglycemia (DOID:4195), validated by many trials like NCT03775733 and NCT01911663, as well as blood pressure-related diseases like essential hypertension (Figure 5c). American ginseng was found to potentially affect hypertension, multiple sclerosis, respiratory infection and upper respiratory tract infections, which were also found in many clinical trials (Figure 5d), while cinnamon potentially intervened diseases related to diabetes or insulin resistance (Figure S22e, Supporting Information) and ginseng potentially affected various diseases, including Alzheimer's disease, atherosclerosis, glucose-related disorders, liver function issues, and multiple sclerosis (Figure S22f, Supporting Information).

3. Discussion

As the demand for precision medicine continues to increase,^[41–45] the intervention of a single drug has gradually become insufficient to achieve perfect precision medicine, while complex systems composed of multiple compounds offer a promising and potential direction for precision medicine.^[46–48] However, despite the accumulation of vast amounts of multi-omics data from single-compound interventions^[49] and the development of numerous deep models focused on single-compound predictions and generation,^[50–52] data reflecting the effects of complex system interventions and models capable of predicting or generating complex system-related functions are scarce. Against this backdrop, there is an urgent need to develop a model capable of predicting the intervention effects of various complex systems, such as NP, on different cell conditions.

Thus, we proposed a model named SETComp, which is capable of predicting genome-wide, cell-specific directed intervention effects for complex systems, such as NP. The two versions of the model (the Concat version with $\approx 200\text{M}$ parameters and the Add version with $\approx 173\text{M}$ parameters) are based on transfer learning and permutation-invariance, with pre-trained compounds as single-element set representations from 970481750 compound-cell-gene associations, and further fine-tuned on 2579488 compound-cell-gene associations derived from GEO data and literature. The model achieved an accuracy of 93.86% and 92.70%, AUC of 0.9888 and 0.9856, respectively, on the test set of complex system-cell-gene associations. In the ablation experiments, we demonstrated the improvement in model prediction performance brought by permutation-invariance, and in tasks involving complex systems that the prediction model had never encountered before, we achieved prediction accuracies of 82.75% and 82.66%. We also explored the relationship between the model's predicted output and the true fold change, and found a strong correlation between them. This provides a new perspective for the subsequent development of quantitative prediction or generation models. The model was further validated through in vitro real-world transcriptomic assays that we personally conducted, achieving an accuracy of up to 88.65% among the intervention of multiple NP. We further demonstrated SETComp's

potential in various biomedical application scenarios, including mechanism uncovering and repositioning of NP.

There were still some limitations in our studies, including the lack of consideration for the quality ratio of each compound in the complex system, and the inability to achieve quantitative prediction of complex system-cell-gene associations. Fortunately, the set embedding module based on set-based composition, compared to directly summing the feature coefficients of the compounds that make up the complex system (which assumes each compound has the same proportion), can, to some extent, address this issue through various attention mechanisms. Regarding quantitative prediction, although we have not directly achieved quantitative prediction of complex system-cell-gene associations, we have deeply explored the relationship between the model's predicted output and the actual fold change, observing a strong positive correlation. This provides a promising perspective for the future development of quantitative prediction or generation models.

For future research, on one hand, we plan to collect more single-cell level compound intervention effect data based on the recently published Tahoe-100M dataset,^[49] combined with the current 180M-level cell-line-based compound intervention expression profiles, for pre-training generative models. On the other hand, we are currently conducting single-cell sequencing experiments on the intervention effects of multiple complex systems in animals to provide data from a single-cell perspective, which will be used to fine-tune the generative model, ultimately achieving the generation of cell-specific whole-genome expression profiles after complex system interventions. We believe that such a generative model for cell-specific whole-genome expression profiles of complex system intervention will provide better basis for prediction in multiple biomedical application scenarios, such as complex system mechanism uncovering, and drug repositioning.

4. Experimental Section

Construction and Training of the SETComp Model—Data Collection, Pre-process, and Generation: In this work, we collected transcriptomics data of various compounds on different cell lines in CMap from LINCS in NIH Common Fund program,^[16] preprocessing with python package cmapPy v4.0.1, and the data was then enhanced using a modified CycleGAN model^[19] to improve the feature dimension from the original 978 to the genome-wide 23614. Consider the mapping function $G: X \rightarrow Y$, where X represents the original L1000 assay and Y denotes the inferred RNA-seq assay, along with its corresponding discriminator D_Y . Accordingly, the adversarial loss function is defined as:

$$L_{CAN}(G, D_Y, X, Y) = E_{y \sim p_{data}(y)} [D(y)]^2 + E_{x \sim p_{data}(x)} [1 - D(G(x))]^2 \quad (1)$$

Finally, 1805898 samples with a single intervention of 39321 compounds were used for further study. Compounds with at least 3 technical replications in one cell-line were kept. And the compounds in the processed matrix were annotated in PubChem database^[21] and the chemical structures of these annotated compounds were obtained, represented by SMILES. NP were collected in HERB (v2.0)^[22] database, and information like chemical compositions and Latin names were collected. Compounds in these NP were also annotated in PubChem database for standard PubChem CID and chemical structures. A total of 46419 compounds remained, among which 25751 served as chemical compositions in at least one of 6198 NP.

For compounds, the enhanced transcriptomic data uniformly after 24h of intervention at the highest concentration of the compound was uniformly used to calculate expression pattern of each gene, while transcriptomic data or microarray in which cell lines were intervened by NP were collected from GEO database (update to 2024 May) and literature.^[53,54] The expression data was further processed with limma^[55] and DESeq2^[56] to calculate DEGs with statistical models. The expression pattern of each gene in different cell lines under different interventions by compounds or NP was classified into 3 categories, including Up (log2FoldChange > 0, p value < 0.05), Down (log2FoldChange < 0, p value < 0.05), and No effect (p value > 0.05).

The data and information of cell lines were collected from CCLE and TCGA database. The expressions matrix of CCLE and TCGA were processed with log transform, if necessary. The data and information of genes were collected from STRING database,^[20] including the interactions between two genes and the corresponding encoded protein sequence. Finally, 970481750 compound-cell line-gene pairs and 2579488 NP-cell line-gene pairs were obtained for further data pre-training and fine-tuning in 3 classes (up-regulated, down-regulated, and no change), respectively.

Feature Extraction for Compounds, Genes, and Cell Lines: Based on the processed data, every compound was first embedded in a self-supervised pre-training model, Infograph,^[57] which proposed to maximize the mutual information between the graph-level and node-level representations and was implemented in TorchDrug. In the InfoGraph model, the mutual information estimator $I_{\phi, \psi}$ is modeled using the discriminator T_{ψ} , which is parameterized by a neural network with parameters ψ :

$$I_{\phi, \psi} \left(h_{\phi}^i(G); H_{\phi}(G) \right) = E_{\mathbb{P}} \left[-sp \left(-T_{\phi, \psi} \left(\tilde{h}_{\phi}^i(x), H_{\phi}(x) \right) \right) \right] - E_{\mathbb{P} \times \tilde{\mathbb{P}}} \left[sp \left(T_{\phi, \psi} \left(\tilde{h}_{\phi}^i(x'), H_{\phi}(x) \right) \right) \right] \quad (2)$$

Here, x denotes an input sample, while the negative sample x' is drawn from the distribution $\tilde{\mathbb{P}} = \mathbb{P}$, which matches the empirical distribution of the input space. Additionally, the softplus function is defined as $sp(z) = \log(1 + e^z)$. And the InfoGraph model was trained on the ZINC 2M database using the recommended parameters, with five hidden layers and 300 neurons in each layer. Another embedding for compounds was based on the PubChem fingerprint calculation implemented with scijava package in Python. Finally, each compound was represented by a 1181-D feature vector, consisting of an 881-D fingerprint embedding and a 300-D Infograph embedding. The features of NP were composed of the corresponding features of chemical compositions and assembled into a set for each natural product.

Genes were also embedded in two ways, including Node2Vec based on network relationships and ProtFlash,^[58] A lightweight protein language model, based on corresponding encoded protein sequence. The loss function of the masked training of ProtFlash is defined as:

$$L_{MLM} = E_{s \sim S} E_M \sum_{i \in M} -\log p(s_i | s_M) \quad (3)$$

In which s_i is the true amino acid and s_M is the masked sequence as context. Node2Vec algorithm was implemented by torch geometric to achieve parallel training on GPU and every gene was embedded into 256 D, while ProtFlash model was used with pre-trained weights and every gene was embedded into 768 D.

For cell line embedding, a Variational Autoencoder (VAE) model was first trained on the TCGA expression matrix, and the CCLE expression matrix was then embedded with the trained VAE model. The loss of the VAE model is defined as the sum of MSE loss and KL loss:

$$L_{VAE} = L_{MSE} + L_{KL} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2 + \frac{\beta}{2} \sum_{j=1}^d \left(\sigma_{ij}^2 + \mu_{ij}^2 - \log \sigma_{ij}^2 - 1 \right) \quad (4)$$

In this function, N represents the total number of samples in the dataset, where each sample x_i is the original expression profile and $\hat{x}_i$ is

the corresponding reconstructed profile generated by the decoder. For each sample, the approximate posterior distribution is characterized by the mean μ_{ij} and standard deviation σ_{ij} for the j -th dimension of the latent variable z , with d denoting the overall dimensionality of the latent space. A hyperparameter β is introduced to control the weight of the KL divergence term, allowing for a balance between reconstruction fidelity and latent space regularization.

Grid search was applied to find the best hyper-parameter combination, in which the search ranges were set as follows: batch size: 32, 64, 128, 256; learning rate: 1e-3, 1e-4, 1e-5; dimension in MLP layer 1: 4096, 2048, 1024; dimension in MLP layer 2: 1024, 512, 256; dimension in MLP layer 3: 256, 128, 64; dimension in latent layer: 64, 32, 16. For each combination, the model was trained for 20 epochs. And the loss function was composed of two parts: Mean Squared Error loss as the reconstruction loss and Kullback-Leibler Divergence loss. Finally, every cell line was embedded into 64 D, in which VAE showed the best reconstruction performance.

Main Architecture of the Model: Apart from the feature extraction modules for compounds, genes and cell lines, the main architecture of the model was constructed with three modules including the set embedding module, the attention module and the prediction module. In the set embedding module, we utilized Deep Sets^[17] and Set Transformer^[18] (See Supplementary materials) to effectively model set-structured data inherent in our problem domain. The set embedding module consisted of two Deep Sets models and one Set transformer model, both implemented in Pytorch environment (<https://github.com/juho-lee/set-transformer/tree/master>). Traditional neural network architectures often assume a fixed-size input or a specific ordering of elements, which is unsuitable for sets that are inherently unordered and variable in size. Deep Sets address this challenge by providing a framework that is permutation-invariant to the input set elements. For the Concat version and Add version of the SETComp model, the outputs of the set embedding are as follows, respectively:

$$Set_{Concat} = \text{concaten}(\text{DeepSets}(x), \text{SetTransformer}(\text{DeepSets}(x))) \quad (5)$$

$$Set_{Add} = \text{DeepSets}(x) + \text{SetTransformer}(\text{DeepSets}(x)) \quad (6)$$

In addition, the attention module further refines the representations obtained from the concatenate of the set embedding module for compounds or NP and the extracted gene embedding and cell line embedding. It utilizes advanced attention mechanisms to focus on the most relevant features within the data, enhancing the model's predictive capabilities by capturing intricate patterns and dependencies. Finally, the prediction module takes the refined representations from the attention module to perform the final prediction task. This module typically consists of three fully connected layers that map the high-dimensional embeddings to the desired output space, enabling the model to generate accurate predictions based on the learned representations.

In Vitro Cell Lines Intervention by Multiple NP for Validation—Cell Culture and Related Reagents: The human-derived breast cancer cell line MCF-7 (RRID: CVCL_0031) and the human-derived lung cancer cell line A549 (RRID: CVCL_0023) were both purchased from the China Infrastructure of Cell Line Resources (China) in January 2020. Both cell lines were tested for mycoplasma contamination and confirmed to be mycoplasma-free. These cells were then cultured in DMEM and RPMI-1640 media, respectively, with 10% (v/v) FBS and 100 U/ml streptomycin/penicillin at 37 °C in a 5% CO₂ environment. The Chinese medicinal herbs Astragalus, Cinnamon, and Codonopsis were purchased from Jiangyin Tianjiang Pharmaceutical Co., Ltd. (China). All the herbal granules were stored in a dry, light-protected environment.

Cell Treatment and Drug Administration Method: Human-derived breast cancer cells (MCF-7) and human-derived lung cancer cells (A549) were digested with trypsin, resuspended, and 10 μ L of cell suspension (1:1 dilution) was counted using a cell counting plate. The cells were then seeded in a 10 cm cell culture dish at a density of 4×10^6 cells per well, and placed in the cell incubator to allow attachment. The following day,

after the cells attached, the culture medium was replaced with 8 mL of fresh serum-free medium. Astragalus, Cinnamon, and Codonopsis were adjusted to final concentrations of 200 µg/mL, respectively, and then cultured in a 37 °C, 5% CO₂ cell incubator for 24 h. On the third day, cells from each group were collected into 1.5 mL sterile, enzyme-free, Eppendorf tubes using Trizol reagent, and stored at -80 °C for sequencing.

RNA Extraction, Library Construction, and Sequencing: Total RNA was extracted from the cell lines using TRIzol reagent (Magen). The A260/A280 absorbance ratio of the RNA samples was measured using a Nanodrop ND-2000 (Thermo Scientific, USA), and the RNA Integrity Number (RIN) was determined using an Agilent Bioanalyzer 4150 (Agilent Technologies, CA, USA). Only RNA samples that passed quality control were used for library construction. The PE library was prepared according to the instructions of the ABclonal mRNA-seq Lib Prep Kit (ABclonal, China). mRNA was purified from 1 µg of total RNA using oligo(dT) magnetic beads, and then fragmented in the ABclonal First Strand Synthesis Reaction Buffer. Subsequently, mRNA fragments were used as templates to synthesize the first strand of cDNA using random primers and reverse transcriptase (RNase H). The second strand of cDNA was synthesized using DNA polymerase I, RNase H, buffers, and dNTPs. The synthesized double-strand cDNA fragments were ligated with adapter sequences for PCR amplification. The PCR products were purified and the library quality was assessed using an Agilent Bioanalyzer 4150. Finally, sequencing was performed on the Illumina Novaseq 6000 / MGISEQ-T7 sequencing platforms.

Application of the SETComp Model in Biomedical Scenarios—Mechanism Uncovering for NP: For the prediction of complex system intervention effects, genes were selected from class 0 and 1 with softmax scores greater than or equal to 0, 0.7, and 0.9 as potential upregulated or downregulated genes under different thresholds. These genes were then sorted by their scores (with negative scores assigned to genes predicted as class 1) for subsequent enrichment analysis. Next, we performed GSEA analysis using the R package clusterProfiler (v4.9.1) to identify pathways potentially intervened by the model, and classified the pathways into activation or inhibition based on Normalized Enrichment Score (NES). This enrichment analysis was repeated for potential intervention genes under different thresholds to obtain mechanism uncovering results for each threshold. For visualizing specific pathways, we used the gseaplot2() function from the R package enrichplot.

Repositioning for NP with Clinical Evidences: We collected clinical reports of various complex systems from the HERB (v2.0) database to validate our findings in drug repositioning. Based on the predicted effects of complex system interventions, we selected genes with softmax scores greater than or equal to 0.7 as potential intervention genes. Using the R package clusterProfiler (v4.9.1), we performed KEGG enrichment analysis and DO enrichment analysis on the potential intervention genes to identify diseases or disease-related terms potentially intervened by each complex system. These diseases or disease-related terms were then modularized according to KEGG classifications. We matched the disease or disease-related terms with clinical diseases and observed whether there were relevant clinical trial reports. Sankey diagrams were visualized using the R package ggalluvial (v0.12.5).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

S.L. contributed to the conception and design of the work. B.W. developed the concept of the work, contributed to the design and implementation of the algorithm, and drafted the initial version of the manuscript. P.Y. contributed to conducting experimental validation. B.W., T.Z., and Q.L. contributed to the data collection and preprocessing.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

complex systems, intervention effect prediction, permutation-invariance, transcriptomics, transfer learning

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